

European Patent Office
80298
Munich
Germany

24 August 2023

Our Ref: NODT-004/N01EP 330150-2065

Dear European Patent Office

Re. European patent application no. 18826609.2
SULPHONYL UREA DERIVATIVES AS NLRP3 INFLAMMASOME MODULATORS
NodThera Limited

We refer to the communication pursuant to Article 94(3) EPC dated 17 April 2023 and to the above-identified application.

We lodge herewith an amended set of claims to be substituted for the claims as presently on file. For the convenience of the Examiner, a further copy of the amended claims showing the amendments in track-changes format is also enclosed.

The applicant respectfully requests that any amendment of the description be deferred until such time as an allowable claim set has been agreed upon.

Amendments

Claim 1 has been amended to specify that the C₅-C₆ aryl of R₁ is a phenyl substituted with two or three R₁₅. This amendment finds basis in paragraphs [0151] and [0153] of the application as filed.

Claim 1 has been further amended to specify that at least one R₁₅ is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo. This amendment finds basis in paragraph [0177] of the application as filed.

Claim 1 has also been amended to specify that R₂₅ is a 4- to 8-membered monocyclic heterocycloalkyl. This amendment finds basis in paragraph [045] of the application as filed.

Finally, claim 1 has been amended to fix the antecedent basis of “the C₃-C₈ monocyclic cycloalkyl, polycyclic cycloalkyl” by specifying that the C₃-C₁₆ cycloalkyl is a C₃-C₈ monocyclic cycloalkyl or polycyclic cycloalkyl.

Claims 2, 4, and 9 have been amended to bring them into line with amended claim 1.

The claims are otherwise unamended.

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24 August 2023

Page Two

Inventive step

The Examiner has maintained their objection that the subject matter of the claims as on file lacks an inventive step over D1. This is because the Examiner appears to consider some of the exemplified compounds not to display the improvement evidenced by the comparative data discussed as part of our previous response of 07 October 2022. In particular, the Examiner has highlighted compounds 58, 71, and 90 as having an IC_{50} in the range of 10 to 50 μM .

As set out above, claim 1 has been amended more closely into line with the exemplified compounds which provide IC_{50} values $<10 \mu M$ – i.e. superior to the $> 10 \mu M$ value of the comparative compound of D1. All of the compounds which are exemplified in the application as filed which have IC_{50} values $<10 \mu M$ fall within the scope of amended claim 1. All of the exemplified compounds which have IC_{50} values $>10 \mu M$ have been excluded from the scope of amended claim 1. This includes each of compounds 58, 71, and 90, which were specifically identified by the Examiner.

Accordingly, the previously-discussed comparative data supports the claimed scope of amended claim 1. Furthermore, given the IC_{50} values for all exemplified compounds within the scope of amended claim 1, it is credible that compounds throughout the scope of the claim would possess the claimed improvement.

As such, the subject matter of amended claim 1 involves an inventive step and it is requested that this objection be withdrawn. The remaining claims either depend upon claim 1 or incorporate its subject matter by reference and therefore likewise involve an inventive step at least by virtue thereof.

The Applicant submits that this response overcomes all objections raised in the Examination Report. If, however, the Examiner disagrees, they are requested to issue another Official Communication, or to contact the undersigned by telephone for any minor outstanding matters. Purely as a precaution, should the Examiner be minded to refuse this application, oral proceedings via videoconference are requested under Article 116 EPC before any such decision is taken.

Yours faithfully



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For and on behalf of Cooley (UK) LLP
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